CS2601 Linear and Convex Optimization

Homework 2

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1. Suppose $\boldsymbol{x}, \boldsymbol{y} \in f^{-1}(C)$, then $f(\boldsymbol{x}) \in C, f(\boldsymbol{y}) \in C$
   $\because C$ is convex
   $\therefore \forall \theta \in [1,0], \theta f(\boldsymbol{x}) + \overline{\theta} f(\boldsymbol{y}) = \theta(\boldsymbol{Ax}+\boldsymbol{b}) + \overline{\theta}(\boldsymbol{Ay}+\boldsymbol{b}) = \boldsymbol{A}(\theta\boldsymbol{x}+\overline{\theta}\boldsymbol{y}) + \boldsymbol{b} = f(\theta\boldsymbol{x}+\overline{\theta}\boldsymbol{y}) \in C$
   $\therefore \theta\boldsymbol{x}+\overline{\theta}\boldsymbol{y} \in f^{-1}(C)$
   $\therefore$ If C is convex, so is $f^{-1}(C)$

2. (a).Suppose $\boldsymbol{x}, \boldsymbol{y} \in C$
   $\therefore \exists\{x_i\} \in S, \boldsymbol{x} = \sum_{i=1}^m \theta_i \boldsymbol{x_i}, \theta_i \geq 0, \sum_{i=1}^m \theta_i = 1.$
   $\exists\{x_j\} \in S, \boldsymbol{y} = \sum_{j=1}^n \theta_j \boldsymbol{x_j}, \theta_j \geq 0, \sum_{j=1}^m \theta_j = 1$
   $\forall\theta \in [1,0], \theta\boldsymbol{x}+\overline{\theta}\boldsymbol{y} = \theta\sum_{i=1}^m \theta_i\boldsymbol{x_i} + \overline{\theta}\sum_{j=1}^n \theta_j\boldsymbol{x_j} = \sum_{k=1}^{m+n}\theta_k\boldsymbol{x_k}, \sum_{k=1}^{m+n}\theta_k = \sum_{i=1}^m \theta\theta_i + \sum_{j=1}^n \overline{\theta}\theta_j = \theta+\overline{\theta} = 1, \boldsymbol{x_k} \in S, \theta_k \geq 0$
   $\therefore \theta\boldsymbol{x}+\overline{\theta}\boldsymbol{y} \in C$
   $\therefore C$ is convex
   (b).Since conv S is convex, it contains all convex combinations of points in S, so C   conv S. Since C is convex, C = conv S by the minimality of conv S.

3. Let $V_i = \{\boldsymbol{x} : ||\boldsymbol{x}-\boldsymbol{x_0}||_2 \leq ||\boldsymbol{x}-\boldsymbol{x_i}||_2\}$
   $\forall\boldsymbol{x} \in V_i, ||\boldsymbol{x}-\boldsymbol{x_0}||_2^2 \leq ||\boldsymbol{x}-\boldsymbol{x_i}||_2^2$,that is $(\boldsymbol{x}-\boldsymbol{x_0})^T(\boldsymbol{x}-\boldsymbol{x_0}) \leq (\boldsymbol{x}-\boldsymbol{x_i})^T(\boldsymbol{x}-\boldsymbol{x_i})$
   $\therefore (\boldsymbol{x_i^T}-\boldsymbol{x_0^T})\boldsymbol{x} + ((\boldsymbol{x_i}-\boldsymbol{x_0})^T\boldsymbol{x})^T \leq ||\boldsymbol{x_i}||_2^2 - ||\boldsymbol{x_0}||_2^2$
   $\because ((\boldsymbol{x_i}-\boldsymbol{x_0})^T\boldsymbol{x}) \in \boldsymbol{R}$
   $\therefore ((\boldsymbol{x_i}-\boldsymbol{x_0})^T\boldsymbol{x}) = ((\boldsymbol{x_i}-\boldsymbol{x_0})^T\boldsymbol{x})^T$
   $\therefore (\boldsymbol{x_i}-\boldsymbol{x_0})^T\boldsymbol{x} \leq \frac{||\boldsymbol{x_i}||_2^2-||\boldsymbol{x_0}||_2^2}{2}$
   Then $V = \bigcap_i V_i$. Let $\boldsymbol{A} = [(\boldsymbol{x_1}-\boldsymbol{x_0})^T, (\boldsymbol{x_2}-\boldsymbol{x_0})^T, ..., (\boldsymbol{x_K}-\boldsymbol{x_0})^T]^T$,
   $\boldsymbol{b} = [\frac{||\boldsymbol{x_1}||_2^2-||\boldsymbol{x_0}||_2^2}{2}, \frac{||\boldsymbol{x_2}||_2^2-||\boldsymbol{x_0}||_2^2}{2}, ..., \frac{||\boldsymbol{x_K}||_2^2-||\boldsymbol{x_0}||_2^2}{2}]^T$,
   $\therefore V = \{\boldsymbol{x} : \boldsymbol{Ax} \leq \boldsymbol{b}\}$, V is a polyhedron.

4. (a.)$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
   $\therefore \nabla^2 f(\boldsymbol{x}) \succeq \boldsymbol{O}$, f is convex
   (b.)$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{2}{x_1^3x_2} & \frac{1}{x_1^2x_2^2} \\ \frac{1}{x_1^2x_2^2} & \frac{2}{x_1x_2^3} \end{bmatrix}$
   $\therefore$ When $x_1 > 0, x_2 > 0$, $\nabla^2 f(\boldsymbol{x}) \succeq \boldsymbol{O}$, f is convex
   (c.)$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 0 & 2x_2 \\ 2x_2 & 2x_1 \end{bmatrix}$
   $\therefore$ When $x_1 = 1, x_2 = 1$, $\nabla^2 f(x_1, x_2)$ is neither positive semidefinite nor negative semidefinite.
   $\therefore$ f is neither concave nor convex

(d.)$\nabla^2 f(x_1,x_2) = \begin{bmatrix} 0 & -\frac{1}{2}x_2^{-\frac{3}{2}} \\ -\frac{1}{2}x_2^{-\frac{3}{2}} & \frac{3}{4}x_1x_2^{-\frac{5}{2}} \end{bmatrix}$

$\therefore$ When $x_1 = 1, x_2 = 1$, $\nabla^2 f(x_1,x_2)$ is neither positive semidefinite nor negative semidefinite.

$\therefore$ f is neither concave nor convex

(e.)$\nabla^2 f(x_1,x_2) = \begin{bmatrix} \alpha(\alpha-1)x_1^{\alpha-2}x_2^{1-\alpha} & \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \\ \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} & -\alpha(1-\alpha)x_1^{\alpha}x_2^{-\alpha-1} \end{bmatrix}$

If $\alpha = 0, f(x_1,x_2) = x_2$, f is convex

If $\alpha = 1, f(x_1,x_2) = x_1$, f is convex

If $0 < \alpha < 1, \nabla^2 f(x_1,x_2) \preceq \boldsymbol{O}$, f is concave

5. (a.)$\forall \boldsymbol{x}, \boldsymbol{y} \in S_\alpha$

$f(\theta\boldsymbol{x} + \overline{\theta}\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + \overline{\theta} f(\boldsymbol{y}) < \theta\alpha + \overline{\theta}\alpha = \alpha$

$\therefore \theta\boldsymbol{x} + \overline{\theta}\boldsymbol{y} \in S_\alpha, S_\alpha$ is convex

$\forall \boldsymbol{x}, \boldsymbol{y} \in C_\alpha$

$f(\theta\boldsymbol{x} + \overline{\theta}\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + \overline{\theta} f(\boldsymbol{y}) \leq \theta\alpha + \overline{\theta}\alpha = \alpha$

$\therefore \theta\boldsymbol{x} + \overline{\theta}\boldsymbol{y} \in C_\alpha, C_\alpha$ is convex

(b.)$dom f = \{\boldsymbol{x} : \boldsymbol{x} \in \mathbb{R}, f(\boldsymbol{x}) < +\infty\}$

In (a.), let $\alpha = +\infty$, then $dom f = S_\alpha$ is convex.

(c.)Suppose M is not empty. Let α to be the global minima of f over X.

$\because \forall \boldsymbol{x} \in M\ \boldsymbol{x} \in X, f(\boldsymbol{x}) = \alpha$

$\therefore \boldsymbol{x} \in C_\alpha, M \subseteq X, M \subseteq C_\alpha$

$\therefore M \subseteq X \cap C_\alpha$

$\forall \boldsymbol{y} \in X \cap C_\alpha,\ f(\boldsymbol{y}) \leq \alpha \leq f(\boldsymbol{x}), \forall \boldsymbol{x} \in X$

$\therefore \boldsymbol{y} \in M$

$\therefore X \cap C_\alpha \subseteq M$

$\therefore M = X \cap C_\alpha$

$\because X, C_\alpha$ is convex

$\therefore M$ is convex

6. (a.)Suppose c < a < b:

$\because f(x)$ is convex, $\frac{a-c}{b-c} + \frac{b-a}{b-c} = 1, 0 < \frac{a-c}{b-c} < 1, 0 < \frac{b-a}{b-c} < 1$

$\therefore \frac{a-c}{b-c} f(b) + \frac{b-a}{b-c} f(c) \geq f(\frac{a-c}{b-c} \cdot b + \frac{b-a}{b-c} \cdot c) = f(a)$

$\therefore \frac{f(a)-f(c)}{a-c} \leq \frac{f(b)-f(c)}{b-c}$

Suppose a < c < b:

$\because f(x)$ is convex

$\therefore \frac{f(b)-f(c)}{b-c} \geq f'(c) \geq \frac{f(c)-f(a)}{c-a}$

$\therefore \frac{f(a)-f(c)}{a-c} \leq \frac{f(b)-f(c)}{b-c}$

Suppose a < b < c:

$\because f(x)$ is convex, $\frac{b-c}{a-c} + \frac{a-b}{a-c} = 1, 0 < \frac{b-c}{a-c} < 1, 0 < \frac{a-b}{a-c} < 1$

$\therefore \frac{b-c}{a-c} f(a) + \frac{a-b}{a-c} f(c) = \frac{b-c}{a-c}(f(a) - f(c)) + f(c) \geq f(\frac{b-c}{a-c} \cdot a + \frac{a-b}{a-c} \cdot c) = f(b)$

$\therefore \frac{f(a)-f(c)}{a-c} \leq \frac{f(b)-f(c)}{b-c}$

$\therefore$ For distinct a, b, c S and a < b, $\frac{f(a)-f(c)}{a-c} \leq \frac{f(b)-f(c)}{b-c}$

(b.)

$\because, x$ is an interior point of S

$\therefore \exists x_0 \in S, x_0 < x; x_1 \in S, x < x_1$
Let $h(y) = \frac{f(y)-f(x)}{y-x}, y \in (x_0, x)$
Then from (a) we can know that $h(y)$ is increasing, and $\frac{f(x_0)-f(x)}{x_0-x} \leq h(y) \leq \frac{f(x_1)-f(x)}{x_1-x}$, which means $h(y)$ is bounded.
$\therefore lim_{y \to x} h(y) = D_- f(x)$ exists.
Let $t(y) = \frac{f(y)-f(x)}{y-x}, y \in (x, x_1)$
Then from (a) we can know that $t(y)$ is increasing, and $\frac{f(x_0)-f(x)}{x_0-x} \leq t(y) \leq \frac{f(x_1)-f(x)}{x_1-x}$, which means $h(y)$ is bounded.
$\therefore lim_{y \to x} t(y) = D_+ f(x)$ exists.
(c.)$\forall x, y \in [0,1], \theta \in [0,1]$
Suppose $x, y \in (0,1], \theta x + \overline{\theta} y \in (0,1]$
$\therefore g(\theta x + \overline{\theta} y) = 0 = \theta g(x) + \overline{\theta} g(y)$
Suppose $x = 0, y \in (0,1], \overline{\theta} y \in (0,1]$
$g(\theta x + \overline{\theta} y) = g(\overline{\theta} y) = 0 \leq \theta g(x) + \overline{\theta} g(y) = \theta$
Suppose $x = y = 0$. Obviously $g(\theta x + \overline{\theta} y) = \theta g(x) + \overline{\theta} g(y)$
Therefore, $\forall x, y \in [0,1], \theta \in [0,1], g(\theta x + \overline{\theta} y) \leq \theta g(x) + \overline{\theta} g(y)$. So $g(x)$ is convex
$D_+ g(0) = lim_{t \downarrow 0} \frac{f(t)-f(0)}{t} = lim_{t \downarrow 0} \frac{0-1}{t} = -\infty$
$\therefore D_+ g(0)$ does not exist.
This does not contradict (b), because 0 is not an interior point of [0,1]